Consistency/Cut Layer for G2-ZOO

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Summary

Pass 69 adds a named APS-level consistency layer to G2-ZOO. For an AMS/preAPS

$$S = (L, \leq, \Box, \boxtimes, T, \perp),$$

define the iterated consistency tower

$$C_0 = T, \qquad C_{n+1} = \boxtimes C_n.$$

The new statement names are:

- $\mathrm{Con}_n^{\mathrm{orb}}$: $C_n \not\leq \perp$.
- $\mathrm{G2}_n$: $C_n \leq \perp \Rightarrow T \leq \perp$.
- $\mathrm{FG2}_n$: $C_{n+1} \leq C_n$, equal to nFG2(n).
- $\mathrm{Flat}_{\leq N}$: distinct values among $C_0, \dots, C_N$ are pairwise incomparable.
- `CutA3`: $x \leq \Box y$ and $x \leq \boxtimes y$ imply $x \leq \boxtimes T$.

`CutA3` is APS axiom A3 read as a cut/collision consistency principle.

Machine-Checked Families

The checker `code/scripts/check-pass69.py` verifies two families.

First, the cycle models C_m for $2 \leq m \leq 12$ are genuine APS models. Their middle orbit is an antichain cycle, so the consistency tower is flat at the level of distinct orbit values. They have no syntactic $\boxtimes$-fixed point, G2 holds vacuously, and every checked nFG2(k) fails.

Second, the detached Rosser period models R_{2k} for $1 \leq k \leq 6$ add a middle atom p with $\boxtimes p = p$. They preserve A1, A2, A4, G2, and the flat consistency orbit, but fail A3 exactly at the detached collision $x = y = p$:

$$p \leq \Box p, \qquad p \leq \boxtimes p, \qquad p \not\leq \boxtimes T.$$

Thus primitive refutability fixed points do not by themselves imply formalized descent or cut/collision closure.

Open Arithmetic Lift

The order-theoretic certificate should not be read as an arithmetic theorem. The next tasks are:

1. Interpret C_n and $\mathrm{Con}_n^{\mathrm{orb}}$ inside $ConLat_T$.
2. Identify the arithmetic counterpart of `CutA3`, comparing Con_T^S, Con_T^H, Rosser consistency, local reflection, and cut admissibility.
3. Decide whether residuation, integrality, and contraction force the detached Rosser fixed point of R_{2k} back into the consistency orbit.
4. Separately return to the Pass-68 problem of comparing $\varprojlim^1(N_n\mathbb{Z})$ with the recollement class ϵ.

Detailed note: `research/notes/g2-zoo-consistency-cut-infinite-pass69.md`

Checker report: `artifacts/reports/pass69-consistency-cut-infinite-g2-zoo-check.json`